

SIOC 217C: Climate and Climate Change

Homework #2

Consider an atmosphere filled by an absorber with uniform mixing ratio that absorbs equally at all terrestrial wavelengths. The optical thickness for this absorber is zero at the top of the atmosphere and has value τ_0 at the surface. For simplicity, assume that atmospheric pressure and density vary exponentially with constant scale height H and ignore any variation of scale height with temperature. Assume the surface is a blackbody and the atmosphere is a graybody (e.g., emits like a blackbody scaled by a fraction equal to emissivity) at terrestrial wavelengths. Also assume that optical thickness can be multiplied by a constant “diffusivity factor” r to account for radiation that goes at all angles of the hemisphere and not just in the vertical direction. For this assignment, it will be most convenient to create a model with layers that have equal thickness in pressure coordinates. Let the surface pressure be p_0 .

It will not be possible to match all characteristics of Earth’s atmosphere and climate with a gray radiative transfer model, so we will choose parameters that match a simple single-layer atmosphere. First, we will assume that the atmosphere absorbs zero solar radiation flux and that solar radiation absorbed by the surface can be expressed as $F_{SW} = (S_0 / 4) (1 - \alpha_p)$ where S_0 is insolation and α_p is the fraction of solar radiation reflected back to space by the surface and atmosphere. A value of $F_{SW} = -239.4 \text{ W m}^{-2}$ (flux defined negative downwards) corresponds to an effective blackbody emission temperature $T_e = 254.9 \text{ K}$. This value of F_{SW} combined with a single-layer greenhouse atmosphere with emissivity $\epsilon_a = 0.772$ above a blackbody surface will produce a radiative equilibrium surface temperature of $T_s = 288.0 \text{ K}$, which is approximately the pre-industrial global mean surface temperature. The temperature of the single-layer atmosphere will be $T_a = 2^{-1/4} T_s = 242.2 \text{ K}$. A total atmospheric optical thickness of $\tau_0 = 0.89$ combined with $r = 1.66$ yields a column atmospheric absorptivity equal to 0.772.

1. a) Let $\tau(p)$ be the value of optical thickness at pressure p integrated downward from the top of the atmosphere, with a value of τ_0 at the surface. Provide an expression for $\tau(p)$ as a function of τ_0 , p , and any other necessary parameters.
b) Plot how $\tau(p)$ varies with p in the range $0 \leq p \leq p_0$ for $\tau_0 = 0.89$.
2. a) Consider a thin layer of the atmosphere at pressure p that has thickness Δp in pressure coordinates and optical thickness $\Delta\tau$, where $\Delta\tau \ll 1$. Let $\epsilon(p)$ be the emissivity of the layer. Provide an expression for $\epsilon(p)$ as a function of Δp , r , and any other necessary parameters.
b) Plot how $\epsilon(p)$ varies with p in the range $0 \leq p \leq p_0$ for $r = 1.66$ and $\tau_0 = 0.89$. Specify what value of $\Delta p / p_0$ you used.
3. a) Let B_s be the amount of broadband hemispheric flux (irradiance) emitted upward by the surface, which can be approximated as a blackbody at terrestrial wavelengths. Calculate the value of B_s if surface temperature T_s is 288.0 K.

- b) Let $F_s(p)$ be the amount of flux emitted upward by the surface that is transmitted to pressure level p . Plot how $F_s(p)$ varies with p in the range $0 \leq p \leq p_0$ for $T_s = 288.0$ K, $r = 1.66$, and $\tau_0 = 0.89$ (units W m^{-2}). Let flux be defined as positive upward.
4. a) Calculate the amount of flux emitted upward to space and downward to the surface by the atmospheric column approximated as a single layer with emissivity $\varepsilon_a = 0.772$ if the atmospheric temperature T_a is 242.2 K.
- b) Let $F_a\uparrow(p)$ be the total amount of flux emitted by all layers of the atmosphere below pressure level p that is transmitted upward to pressure level p . Plot how $F_a\uparrow(p)$ varies with p in the range $0 \leq p \leq p_0$ for $T_a = 242.2$ K, $r = 1.66$, and $\tau_0 = 0.89$ (units W m^{-2}). Assume atmospheric temperature is constant with height. Let flux be defined as positive upward.
- c) Let $F_a\downarrow(p)$ be the total amount of flux emitted by all layers of the atmosphere above pressure level p that is transmitted downward to pressure level p . Plot how $F_a\downarrow(p)$ varies with p in the range $0 \leq p \leq p_0$ for $T_a = 242.2$ K, $r = 1.66$, and $\tau_0 = 0.89$ (units W m^{-2}). Assume atmospheric temperature is constant with height. Let flux be defined as negative downward.
- d) Let $F_a(p) = F_a\uparrow(p) + F_a\downarrow(p)$ be the net flux at each pressure level p that corresponds to the difference in magnitude between the total flux emitted and transmitted upward from layers of the atmosphere below and the total flux emitted and transmitted downward from layers of the atmosphere above. Plot how $F_a(p)$ varies with p in the range $0 \leq p \leq p_0$ for $T_a = 242.2$ K, $r = 1.66$, and $\tau_0 = 0.89$ (units W m^{-2}). Assume atmospheric temperature is constant with height. Let flux be defined as positive upward.
- e) What is the value of F_a at $p = 0$? What is the value of F_a at $p = p_0$? Taking into account discretization error, do these values correspond to what you calculated in Question 4(a)? What change in parameter or assumption would cause the answers to this question and Question 4(a) to be different?
5. a) Let $F_{LW}(p) = F_s(p) + F_a(p)$ be the net radiation flux at each pressure level p that corresponds to the difference in magnitude between the total flux emitted and transmitted upward from the surface and layers of the atmosphere below and the total flux emitted and transmitted downward from layers of the atmosphere above. Plot how $F_{LW}(p)$ varies with p in the range $0 \leq p \leq p_0$ for $T_s = 288.0$ K, $T_a = 242.2$ K, $r = 1.66$, and $\tau_0 = 0.89$ (units W m^{-2}). Assume atmospheric temperature is constant with height. Let flux be defined as positive upward.

- b) In Hartmann Eq. 3.38, dF/dz corresponds to the divergence of the net radiation flux. Since pressure and height increase in opposite directions, $-dF/dp$ will correspond to the divergence of net radiation flux in pressure coordinates. If net flux entering the top and/or bottom of an atmospheric layer is less than net flux exiting the top and/or bottom of the layer, there will be divergence of radiative energy and $-dF/dp$ will be positive. If net flux entering the top and/or bottom of the atmospheric layer is greater than net flux exiting the top and/or bottom of the layer, there will be convergence of radiative energy and $-dF/dp$ will be negative. Where is there divergence of net flux and where is there convergence of net flux in the plot in Question 5(a)? Briefly explain your reasoning.
- c) Let $-dF/dp$ be the divergence of net radiation flux across an atmospheric layer. Plot how $-dF/dp$ varies with p in the range $0 \leq p \leq p_0$ for $T_s = 288.0$ K, $T_a = 242.2$ K, $r = 1.66$, and $\tau_0 = 0.89$ (units $\text{W m}^{-2} \text{Pa}^{-1}$). Assume atmospheric temperature is constant with height. Let flux be defined as positive upward.
- d) Where in the plot in Question 5(c) is there likely to be radiative heating? Where in the plot in Question 5(c) is there likely to be radiative cooling? Briefly explain your reasoning.
- e) Let dT/dt be the rate of temperature change in a layer in the atmosphere. Provide an expression for dT/dt in terms of the divergence of net radiation flux $-dF/dp$ and any other necessary parameters.
- f) Let dT/dt be the rate of temperature change in a layer in the atmosphere. Plot how dT/dt varies with p in the range $0 \leq p \leq p_0$ for $T_s = 288.0$ K, $T_a = 242.2$ K, $r = 1.66$, $\tau_0 = 0.89$, $g_0 = 9.81 \text{ m s}^{-2}$, and $c_p = 1004 \text{ J kg}^{-1} \text{K}^{-1}$ (units K day^{-1}). Assume atmospheric temperature is constant with height.
- g) Let $T(t_0)$ be the temperature at time t_0 and let Δt be an increment of time. Provide an approximate expression for the temperature at time $t_0 + \Delta t$ in terms of the rate of temperature change dT/dt and any other necessary parameters.
- h) With repeated iteration of temperature change over time increments, the atmosphere will converge towards a temperature profile corresponding to radiative equilibrium of the atmosphere. Calculate and plot how atmosphere-only radiative equilibrium temperature $T_{a \text{ only, eq}}(p)$ varies with p in the range $0 \leq p \leq p_0$ for surface temperature $T_s = 288.0$ K that does not change with time, $r = 1.66$, and $\tau_0 = 0.89$.
- i) What is the atmosphere-only radiative equilibrium temperature at $p/p_0 = 0.5$?

A complete radiative equilibrium atmospheric temperature profile can only be obtained by allowing the surface temperature to come to radiative equilibrium. This will require specifying the solar radiation flux F_{SW} and the thermal inertia of the surface, which we will approximate as a certain depth h of water in which heat is uniformly mixed. For now, we will ignore all effects of water on climate in the model except thermal inertia, and we will assume that there is no turbulent exchange of sensible and latent heat between the atmosphere and water surface.

6. a) Let dT_s/dt be the rate of temperature change of the surface. Provide an expression for dT_s/dt in terms of F_{SW} , $F_{LW}(p_0)$, h , and any other necessary parameters.

- b) With repeated iteration of temperature change over time increments, the surface and atmosphere will converge towards values corresponding to radiative equilibrium. Calculate the radiative equilibrium surface temperature $T_{s,eq}$ for $F_{SW} = -239.4 \text{ W m}^{-2}$, $r = 1.66$, and $\tau_0 = 0.89$.
- c) Plot how the radiative equilibrium temperature of the atmosphere $T_{a,eq}(p)$ varies with p in the range $0 \leq p \leq p_0$ for $F_{SW} = -239.4 \text{ W m}^{-2}$, $r = 1.66$, and $\tau_0 = 0.89$.
- d) Plot how terrestrial net radiation flux $F_{LW}(p)$ varies with p in the range $0 \leq p \leq p_0$ for radiative equilibrium surface and atmospheric temperature for $r = 1.66$, $\tau_0 = 0.89$, and $F_{SW} = -239.4 \text{ W m}^{-2}$. Let flux be defined as positive upward.
- e) Briefly explain why the profile of net radiation flux in Question 6(d) must have the shape that it has and the magnitude that it has.
7. a) Let $T_{a,eq}(z)$ be the radiative equilibrium temperature of the atmosphere in height coordinates. Plot how the radiative equilibrium temperature of the atmosphere $T_{a,eq}(z)$ varies with z in the range $0 \leq z \leq 7H$ for $F_{SW} = -239.4 \text{ W m}^{-2}$, $r = 1.66$, and $\tau_0 = 0.89$.
- b) Let ϵ_{top} be the emissivity of a thin layer near the top of the atmosphere. In radiative equilibrium, the following balance will hold, where T_{top} is the temperature of the thin layer and T_e is the effective blackbody emission temperature of Earth. If you do not understand why this balance holds, talk to me.
- $$2 \epsilon_{top} \sigma T_{top}^4 = \epsilon_{top} \sigma T_e^4$$
- What is the value of $2^{-1/4} T_e$ if $T_e = 254.9 \text{ K}$? What is the temperature of the topmost layer of the atmosphere in radiative equilibrium?
- c) What is the magnitude of the radiative equilibrium lapse rate $\Gamma_{top} = -dT/dz$ across the two topmost layers of the atmosphere (units K km^{-1})? Assume that scale height $H = 8 \text{ km}$, $F_{SW} = -239.4 \text{ W m}^{-2}$, $r = 1.66$, and $\tau_0 = 0.89$. If you do not understand why the lapse rate has this value, talk to me.

We will now apply the model to investigate differences in radiation flux and equilibrium temperature between an atmosphere with $1 \times \text{CO}_2$ concentration ($\tau_0 = 0.89$) and an atmosphere with $2 \times \text{CO}_2$ concentration ($\tau_0 = 0.92$). We will assume that the optical thickness of the atmosphere instantly increases from $\tau_0 = 0.89$ to $\tau_0 = 0.92$ and will first consider radiation flux before the temperature of the surface and the atmosphere has had any time to adjust. Then we will consider the temperature of the surface and the atmosphere after they have reached radiative equilibrium with the new optical thickness.

8. a) Calculate the terrestrial net radiation flux $F_{LW}(p)$ when the surface and atmospheric temperature correspond to radiative equilibrium for $\tau_0 = 0.89$ but emissivity and transmissivity correspond to $\tau_0 = 0.92$. Plot how the combined solar and terrestrial net radiation flux $F_{SW} + F_{LW}(p)$ varies with p in the range $0 \leq p \leq p_0$ for $F_{SW} = -239.4 \text{ W m}^{-2}$ and $r = 1.66$. Let flux be defined as positive upward.

- b) The magnitude of the combined solar and terrestrial net radiation flux at the top of the atmosphere after an instantaneous doubling of CO_2 is the radiative forcing ΔQ . What is the value of $\Delta Q = -(F_{SW} + F_{LW})$ at $p = 0$ after τ_0 instantaneously increases from 0.89 to 0.92?
- c) Now let the surface and atmosphere converge towards values corresponding to a new radiative equilibrium. Calculate the radiative equilibrium surface temperature $T_{s,eq}$ for $F_{SW} = -239.4 \text{ W m}^{-2}$, $r = 1.66$, and $\tau_0 = 0.92$.
- d) The climate sensitivity parameter λ_R scales radiative forcing to the change in equilibrium surface temperature.

$$\Delta T_s = \lambda_R \Delta Q$$

What is the value of λ_R in the model (units K per W m^{-2})? Note that this value substantially underestimates λ_R for Earth's climate because it does not include climate feedbacks.

- e) What is the primary physical process controlling the quantitative value of λ_R ? In other words, how does the change in temperature restore the model back to radiative equilibrium? Why is the change in temperature not smaller or larger than it is?
- f) Plot how the difference between $1\times\text{CO}_2$ concentration and $2\times\text{CO}_2$ concentration in radiative equilibrium temperature of the atmosphere $T_{a,eq}(z)$ varies with z in the range $0 \leq z \leq 7H$ for $F_{SW} = -239.4 \text{ W m}^{-2}$, $r = 1.66$, and $\tau_0 = 0.89$ and $\tau_0 = 0.92$.
- g) Why is there no change in the temperature near the top of the atmosphere?